# Interrogating Alien Life: Adversarially Trained Reinforcement Learning for Substrate-Agnostic Detection of Self-Repair Dynamics

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**Abstract**

Life-detection strategies have largely relied on classifying *static* chemical or spectral patterns. Such strategies are exposed to abiotic false positives and, as recent reports on machine-learning classifiers suggest, to adversarial confusion. We propose instead that life detection be framed as an interactive, adversarial inference problem. A reinforcement-learning agent, the *prober*, selects graded perturbations of a sample (radiation, thermal, redox) and observes the ensuing recovery dynamics. Because maintaining organization against damage is a defining operational feature of living systems, we hypothesize that the dose–response and time–response structure of recovery carries information about active repair that is largely independent of molecular substrate. To keep the test from being fooled by non-living dissipative systems, the prober is trained against an adversarial *impostor generator* that constructs abiotic dynamics under physical admissibility constraints. We formalize the game as a partially observable Markov decision process with a mission budget; show that the value of the game is a total-variation distinguishability between living-like and impostor trajectory laws; and show that adaptive interrogation weakly dominates fixed protocols. We specify simulation testbeds, baselines, metrics, pre-registrable hypotheses with explicit falsification criteria, and a staged wet-laboratory validation plan built around killed-cell and abiotic-dissipative controls. **No empirical results are reported;** the contribution is a framework whose central claims are stated so that they can be refuted.

**Keywords:** life detection; agnostic biosignatures; reinforcement learning; adversarial robustness; self-repair; artificial life; POMDP

## 1 Introduction

The search for life beyond Earth is shaped by an asymmetry. A single confirmed detection would rank among the most consequential findings in the history of science, yet the evidence supporting it is likely to be indirect, ambiguous, and gathered by instruments that are small, remote and budget-limited. The community's response has been to organize evidence into explicit frameworks such as the Ladder of Life Detection [16], and to seek *agnostic* biosignatures that do not presuppose Earth-like biochemistry [2, 7, 8, 14, 22]; see also the community strategy in [19].

Nearly all proposed signatures share one design choice: they are *static*. They ask whether a sample, atmosphere or planetary population shows an anomalous pattern (an abundance distribution, a complexity measure, a spatial correlation) at the moment of observation. Static signatures are attractive because they are passive and cheap, but they carry two weaknesses. First, abiotic processes can produce surprising patterns, so any single anomaly invites a non-biological explanation. Second, when the final judgement is delegated to a learned classifier, the classifier itself becomes an attack surface: small, structured departures from the training distribution can induce confident errors [10]. Recent work presented at ALIFE 2026 is reported to have driven machine-learning life detectors to a 100% false-positive rate with adversarially constructed inputs [4]†.

We argue that the remedy is not only a better classifier but a different experimental stance. A chemist who suspects a sample is alive does not merely look at it; the chemist *does something to it*—feeds it, warms it, irradiates it—and watches what happens. A living system is distinguished less by what it is made of than by

what it does when damaged: it senses the damage, mobilizes resources, repairs, and often becomes better prepared for the next insult. Maintaining organization far from equilibrium requires continual dissipation and repair, and theoretical work on the statistical physics of self-replication ties such organization to thermodynamic constraints rather than to a particular chemistry [9]. The trade-off between investment in maintenance and other demands is, moreover, central to the evolutionary theory of ageing [13], which suggests that repair capacity is a measurable, evolutionarily tuned property of living systems.

This paper proposes **active interrogation of self-repair** as a framework for life detection, and makes it adversarially robust by construction. A reinforcement-learning *prober* chooses perturbations sequentially under a mission budget; a *generator* of abiotic impostors is trained to imitate the recovery behaviour of living-like systems; and the value of the resulting zero-sum game measures how difficult it is for non-living matter to imitate the way living matter responds to being damaged.

**Contributions.**

- **C1 — Problem formulation.** Life detection as a budgeted POMDP over perturbation actions, with recovery-response observables (Section 3).
- **C2 — Adversarial formulation and two propositions.** The game value equals a total-variation distinguishability between living-like and impostor trajectory laws (Proposition 1); adaptive interrogation weakly dominates fixed protocols, with a transparent example where the advantage is exponential (Proposition 2).
- **C3 — A tiered, falsifiable research programme.** Simulation testbeds, baselines, metrics and decision rules (Section 4).
- **C4 — A staged wet-laboratory plan** featuring killed-cell controls that hold biochemical composition fixed while removing repair (Section 4.4).

**Scope statement.** This manuscript reports no experiments. It is a theoretical framework and a registered-style research plan. Its value lies in the clarity of its claims and in how readily they can be refuted; Section 5 enumerates the outcomes that would count against it.

## 2 Background and Related Work

### *2.1 Biosignature frameworks and agnostic signatures*

The Ladder of Life Detection organizes lines of evidence by their strength and by the confidence they provide [16]. Agnostic proposals include assembly-theoretic measures of molecular complexity [14], epsilon-machine measures of planetary complexity [2], monomer-abundance distributions that reflect selection acting on molecular building blocks, demonstrated in terrestrial chemistry and in digital organisms [7, 8], and population-scale correlations produced by life spreading between planets [22]. Benner argues for detecting Darwinian behaviour from molecules in planetary waters [3]. All of these are measurements of a state; none specifies a policy for *acting on* the sample to elicit information.

### *2.2 Decision-making for life-detection missions*

Partially observable Markov decision processes [12] provide a standard language for sequential decisions under uncertainty. A recent preprint applies a POMDP framework to sequence the instruments of the Enceladus Orbilander life-detection suite, with offline-computed policies so that behaviour can be validated before deployment [1]†. That work optimizes *which instrument to run when*; the present proposal adds a new action class, *perturbation of the sample*, and a new adversarial training signal. Bayesian experimental design [5] supplies the information-theoretic vocabulary for selecting such actions.

### 2.3 Adversarial robustness

Adversarial examples show that high accuracy on a training distribution does not imply reliability under structured shifts [10]. Reports of this failure in biosignature classifiers [4]† concern *passive* classifiers operating on fixed data. Our approach differs in that the adversary is part of training from the outset, and the prober can query the sample further rather than being confined to a single snapshot.

### 2.4 Artificial life and simulation testbeds

Artificial-life systems offer ground-truth-labelled testbeds. Examples include continuous cellular automata such as Lenia [6], neural cellular automata trained to grow and regenerate [15], neural cellular automata exhibiting self-replication and mutation [21], and reaction–diffusion systems whose abiotic spots can self-replicate [17]. The last is especially instructive: self-replicating spots are non-living by any ordinary standard, and are therefore natural hard impostors.

## 3 Formal Framework

### 3.1 Samples, perturbations and observations

A *sample* is a controlled stochastic dynamical system with latent state $x_t \in X$. The prober chooses an action $a_t = (m, d, \tau)$ consisting of a perturbation modality m (for example ultraviolet, thermal or oxidative), a dose $d \in [0, d_{max}]$ and a duration $\tau$; it may also choose to wait, or to stop and decide. The instrument returns a noisy observation $y_t$:

$$x_{t+1} \sim P(\cdot \mid x_t, a_t), \qquad y_t \sim O(\cdot \mid x_t) \tag{1}$$

Each sample belongs to a hidden class $z \in \{L, I\}$. Class L (*living-like*) contains self-sustaining systems capable of Darwinian evolution, following the widely used working definition [11]. Class I (*impostor*) contains admissible non-Darwinian dissipative systems, defined in Section 3.4.

### 3.2 Recovery-response observables

Let $\Phi: X \to [0, 1]$ be a functional order parameter, normalized so that $\Phi = 1$ at the unperturbed baseline (for example a fluorescence, redox or structural-coherence index). After a single perturbation of dose d, define the mean recovery curve $\rho(t; d)$ and the damage depth $\delta(d)$:

$$\rho(t; d) = \mathbb{E}[\Phi(x_t) \mid a = (m, d, \tau)], \qquad \delta(d) = 1 - \min_t \rho(t; d) \tag{2}$$

The instantaneous recovery rate and its time average, defined while $\rho < 1$, are

$$r(t; d) = \frac{\dot{\rho}(t; d)}{1 - \rho(t; d)}, \qquad \bar{r}(d) = \frac{1}{T}\int_0^T r(t; d)\, dt \tag{3}$$

We highlight three candidate features, each a *hypothesis* about how repair differs from passive relaxation. (i) A **threshold dose**, beyond which recovery collapses:

$$d_\varepsilon^* = \inf\{\, d : \rho(T; d) < 1 - \varepsilon \,\} \tag{4}$$

(ii) An **induction index** measuring whether the recovery rate increases with damage depth, as when damage switches on repair machinery:

$$\gamma = \frac{d\bar{r}}{d\delta} \tag{5}$$

A system relaxing passively toward a stable attractor is expected to have $\gamma \approx 0$ in the linear regime, whereas induced repair gives $\gamma > 0$. (iii) **Priming**: a second identical dose after an interval elicits a response that differs from the first (adaptive response or hormesis). Figure 2 illustrates these features schematically.

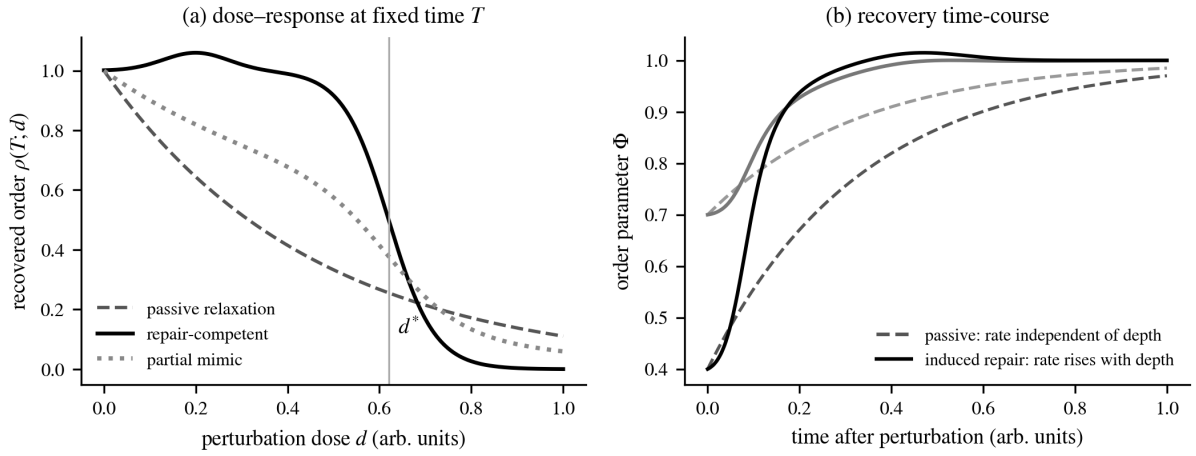



**Figure 2.** Schematic of hypothesized recovery signatures. (a) Dose–response at fixed time: a repair-competent system may hold near-full recovery up to a threshold d* and then collapse, whereas passive relaxation decays smoothly; a partial mimic lies between. (b) Recovery time-course: in passive relaxation the rate is independent of damage depth, whereas induced repair accelerates with depth and may overshoot. *These curves are hypothetical illustrations, not measured data.*

We stress that these features are hypotheses to be tested, not assumptions of the method. The prober policy is trained end-to-end on raw observations; the features serve as interpretable summaries and as inputs to a hand-designed baseline (B3, Section 4.2). If learned policies do no better than these features, or the features fail to separate the classes, the framework should say so.

### *3.3 The prober as a budgeted POMDP*

Perturbations consume sample, time and energy, and may be destructive. We therefore attach a cost c(a) to each action and impose a mission budget B. The history is $h_t = (a_1, y_1, \ldots, a_t, y_t)$, and the prober's belief about the class is

$$b_t(z) = P(z \mid h_t), \qquad \sum_t c(a_t) \le B \tag{6}$$

A recurrent policy $\pi_\theta$ maps histories to actions, including the terminal actions "decide L" and "decide I". The terminal reward is $+1$ for a correct decision and $-1$ otherwise. To speed learning we shape the per-step reward with expected information gain (cf. [5]) and an explicit cost term:

$$r_t = -\lambda\, c(a_t) + \beta\,[H(b_{t-1}) - H(b_t)] \tag{7}$$

where H is Shannon entropy and $\lambda$, $\beta \ge 0$ are weights. Shaping terms are used only for training; evaluation uses terminal accuracy at a stated budget.

### *3.4 The impostor generator and admissibility*

The generator $G_\varphi$ proposes systems from an *admissible* family $\Phi$ of abiotic dynamics. Admissibility constrains the adversary so that a "win" is scientifically meaningful rather than an artefact of an unphysical or secretly living impostor. We require that impostors (a) respect locality and conservation laws appropriate to the substrate, (b) have bounded energy flux, and (c) **lack a mechanism for heritable variation** (no copy operator with mutation), so that they cannot be Darwinian by construction. Within these limits the family is deliberately rich: mass-action reaction networks with bounded rate constants, Gray–Scott-type reaction–diffusion systems [17], and neural cellular automata [15] that may be trained to regenerate by gradient descent. Such designed regenerators are repair-competent but not evolved, which makes them a pointed test of whether recovery behaviour alone discriminates *evolved* life from *engineered* repair.

### *3.5 Game value and two propositions*

Fix a budget-feasible policy $\pi$ and let $P_L^{\pi}$ and $P_{I,\varphi}^{\pi}$ be the laws of the interaction trajectory under classes L and I. With equal priors, a decision rule $\hat{z}$ has accuracy

$$\mathrm{Acc}(\pi, \hat{z}; \phi) = \tfrac{1}{2} P_L^{\pi}(\hat{z} = L) + \tfrac{1}{2} P_{I,\phi}^{\pi}(\hat{z} = I) \tag{8}$$

and we define the *value of the interrogation game* as

$$V^* = \min_{\phi \in \Phi} \max_{\pi \in \Pi_B} \max_{\hat{z}} \mathrm{Acc}(\pi, \hat{z}; \phi) \tag{9}$$

where $\Pi_B$ is the set of policies that respect the budget. **Proposition 1 (value as distinguishability).** For every fixed $\pi$ and $\varphi$,

$$\max_{\hat{z}} \mathrm{Acc}(\pi, \hat{z}; \phi) = \tfrac{1}{2}\left[1 + \mathrm{TV}\left(P_L^{\pi}, P_{I,\phi}^{\pi}\right)\right] \tag{10}$$

*Proof.* The Bayes-optimal rule chooses, for each trajectory $\tau$, the class with the larger probability. Its accuracy is $\tfrac{1}{2} \Sigma_\tau \max(p(\tau), q(\tau))$ with $p = P_L^{\pi}$ and $q = P_{I,\varphi}^{\pi}$. Since $\max(p, q) = (p + q + |p - q|)/2$ and $\Sigma p = \Sigma q = 1$, this sum equals $\tfrac{1}{2}[1 + \mathrm{TV}(p, q)]$ with $\mathrm{TV}(p, q) = \tfrac{1}{2}\Sigma|p - q|$. □

**Corollary.** $V^* = \tfrac{1}{2}$ if and only if some impostor distribution in $\Phi$ (mixtures allowed) reproduces the living-like trajectory law under *every* budget-feasible policy. In that case the interrogation observables carry no information about class within the modelled families, and the test is vacuous. Any $V^* > \tfrac{1}{2}$ is a quantitative lower bound on how well non-living matter, within $\Phi$, can be told apart from living-like matter by interrogation. We do not claim that the game has a saddle point: the objective need not be concave in $\pi$. Training uses alternating best responses, and we report robustness against held-out adversaries (Eq. 11) rather than an equilibrium.

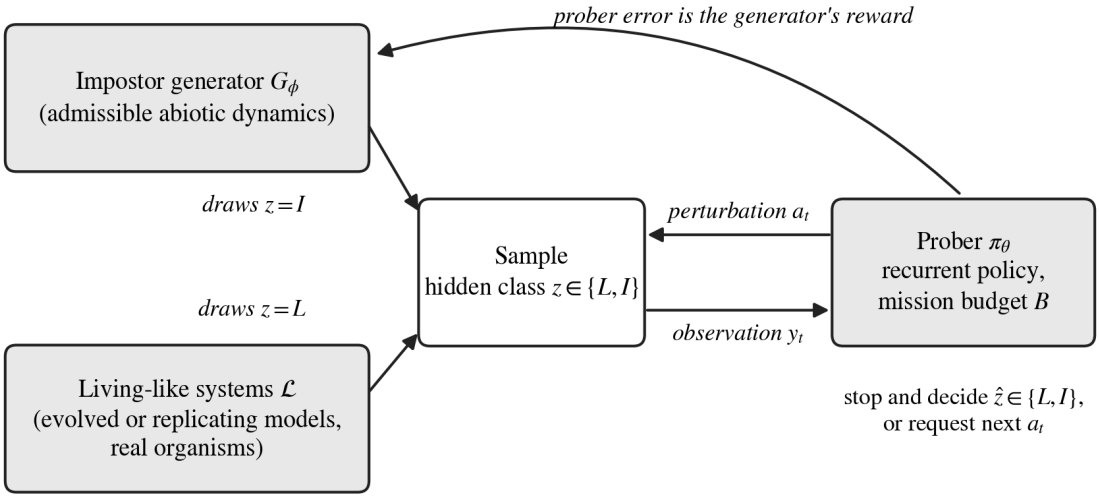
**Proposition 2 (adaptivity weakly dominates).** Let $\Pi_B^{\mathrm{fix}} \subset \Pi_B$ be the policies that commit to a perturbation schedule in advance, and let $V^*_{\mathrm{fix}}$ be the value with $\Pi_B^{\mathrm{fix}}$ in place of $\Pi_B$. Then $V^*_{\mathrm{fix}} \leq V^*$. In particular, passive classification (the policy that performs no perturbation) is a special case, so active interrogation cannot do worse than a static classifier *when perturbation costs are correctly accounted for in B*. *Proof.* Enlarging the policy class cannot decrease a maximum. □

**Example 1 (where adaptivity helps).** Suppose living-like systems exhibit a sharp transition of width w in dose–response, while impostors show a smooth decline of width $w' >> w$, at an unknown location in [0, 1]. Telling these apart requires resolving the transition. In the noise-free idealization, a fixed grid needs on the order of 1/w doses, whereas adaptive bisection, which concentrates doses where the response changes, needs on the order of $\log_2(1/w)$. Real measurements are noisy, so this exponential gap is an upper-end illustration of why adaptive protocols matter under tight budgets, not a forecast.

Finally, we define the headline robustness metric. Let $\Phi_{\mathrm{held}}$ be adversary families never seen in training. The *robust accuracy* of a trained prober is

$$A_{\mathrm{rob}}(\theta) = \min_{\phi' \in \Phi_{\mathrm{held}}} \mathrm{Acc}(\pi_\theta, \hat{z}_\theta; \phi') \tag{11}$$

which is estimated by training fresh adversaries against the frozen prober (a best-response attack) and also by evaluating hand-built adversary families.




Zero-sum training: the prober maximises accuracy per perturbation spent; the generator maximises the prober's error.

**Figure 1.** The interrogation game. A hidden class is drawn; the sample is either a living-like system or an impostor produced by the generator. The prober selects perturbations and observes recovery under a budget, then decides. The prober is rewarded for correct, economical decisions; the generator is rewarded for the prober's errors.

| Symbol | Meaning |
|---|---|
| $z \in \{L, I\}$ | Hidden class: living-like (L) or impostor (I) |
| $a = (m, d, \tau)$ | Perturbation: modality, dose, duration |
| $\Phi$, $\rho(t; d)$ | Functional order parameter; mean recovery curve after dose d |
| $\delta(d)$, $r(t; d)$, $\gamma$ | Damage depth; recovery rate; induction index |
| $d^*_\varepsilon$ | Threshold dose beyond which recovery at time T falls below $1 - \varepsilon$ |
| $\pi_\theta$, B, c(a) | Prober policy; mission budget; action cost |
| $G_\varphi$, $\Phi$ | Impostor generator; admissible impostor family |
| $V^*$, $A_{rob}$ | Game value; robust accuracy against held-out adversaries |

**Table 1.** Notation.

# 4 Research Programme and Experimental Design

## *4.1 Training procedure*

The prober is a recurrent policy trained with proximal policy optimization [20] on the budgeted POMDP of Section 3.3 (see [23] for background on policy-gradient methods). The generator's parameters are updated with evolution strategies [18], which suit non-differentiable simulators and discrete structural choices. To limit cycling between exploiter and defender we keep an archive of past impostors and sample from it during prober updates, in the manner of league training. Appendix A gives the loop.

## *4.2 Simulation testbeds and baselines*

We propose four tiers of increasing realism, so that failures can be localized. In silico perturbations include random deletion of a fraction d of state variables (a radiation analogue), parameter shifts (a thermal analogue) and flux changes (a redox analogue). Observables are a low-dimensional order parameter plus noise.

| Tier | Living-like class (L) | Impostor class (I) | Purpose |
|---|---|---|---|
| 0 | Analytic toy: induced-repair kinetics with threshold | Linear relaxation; smooth saturating decay | Verify the pipeline and Propositions 1–2 |
| 1 | Autocatalytic networks with replication and mutation | Mass-action networks; Gray–Scott patterns incl. self-replicating spots [17] | First realistic test of discrimination |
| 2 | Evolved digital organisms [8]; neural CA with replication and mutation [21] | Lenia [6]; neural CA trained to regenerate [15] | Evolved vs engineered repair |
| 3 | Protocell models with growth, replication and mutation [21] | Adversarially generated dynamics from $\Phi$ | Held-out substrate transfer |

**Table 2.** Simulation tiers. Ground truth is fixed by construction in simulation.

| Baseline | Description |
|---|---|
| B0 | Random perturbation schedule at the same budget |
| B1 | Fixed dose ladder (non-adaptive, equally spaced doses) |
| B2 | Static snapshot classifier on unperturbed observations (the passive baseline) |
| B3 | Bayesian POMDP over hand-designed features d*, γ, priming (Section 3.2) |
| B4 | Learned adaptive prober trained *without* the adversary (isolates the effect of adversarial training) |

**Table 3.** Baselines.

### *4.3 Hypotheses, metrics and decision rules*

Thresholds below are *proposed starting points*, to be fixed before any experiment is run and adjusted only through a documented amendment. All results are reported over at least 20 random seeds with bootstrap 95% confidence intervals.

| Hypothesis | Statement | Proposed decision rule |
|---|---|---|
| H1 | Learned adaptive interrogation beats static and fixed-schedule baselines | Balanced-accuracy gain over the best of B0–B3 of at least 0.05 at $B \leq 10$ perturbations; confidence interval excludes zero |
| H2 | Adversarial training yields robustness | $A_{rob} \geq 0.80$, and $A_{rob}$ of the prober exceeds that of B4 (non-adversarial) by a margin whose interval excludes zero |
| H3 | Discrimination transfers across substrates | Accuracy drop of at most 0.10 when trained on Tiers 1–2 and tested on held-out Tier 3 systems |
| H4 | Signature is observable in real matter | AUROC $\geq 0.90$ for live vs killed cells and $\geq 0.85$ for live cells vs abiotic dissipative systems |

**Table 4.** Pre-registrable hypotheses. Secondary metrics: expected information gain per perturbation, calibration error, accuracy as a function of budget, and which doses and modalities the learned policy favours (interpretability).

### *4.4 Staged wet-laboratory validation*

**Stage A (controlled materials).** Non-pathogenic bacteria handled at the biosafety level appropriate to the strain (for example *Escherichia coli* K-12 or *Bacillus subtilis*), with the radiation-tolerant bacterium *Deinococcus radiodurans* as a positive control for strong repair. The comparison classes are: (i) live cells; (ii) **killed cells** (heat- or UV-killed), which retain biochemical composition but lack repair, making them the critical control that separates *what the sample is made of* from *what it does*; and (iii) abiotic dissipative systems such as Belousov–Zhabotinsky-type oscillators, coacervate droplets and mineral growth systems. Perturbations are graded UV-C, thermal shock and oxidative stress; readouts are low-cost, flight-plausible measurements such as a resazurin redox indicator and optical density, with a nanomotion or fluorescence channel as an optional extension.

**Stage B (environmental samples).** Analogue-site samples (for example permafrost, hot-spring or hypersaline material) alongside sterilized splits of the same material. Contamination control is essential: a positive result must be distinguishable from terrestrial contamination of the apparatus. Because perturbations are destructive, each action consumes an aliquot, so the budget B has a physical meaning in this stage.

### *4.5 Relation to mission operations*

A trained prober is a small recurrent network whose inference cost is modest, and its policy can be precomputed offline and validated before launch, in the same spirit as POMDP-based instrument sequencing [1]†. In a mission, the perturbation actions would be an additional action class beside instrument scheduling, with sample aliquots as the budgeted resource. We make no claim that such an instrument exists or is flight-ready; the point is that the policy formalism already used for mission autonomy extends naturally to interrogation.

## 5 Anticipated Outcomes and Falsification

Every outcome of the programme is informative, and the framework is falsifiable at several points. Table 5 maps outcomes to conclusions.

| Outcome | Interpretation |
|---|---|
| O1. H1–H3 supported in simulation; H4 supported in the laboratory | Recovery-response interrogation is a promising complementary biosignature; proceed to analogue-site tests |
| O2. Separation without the adversary, collapse against held-out adversaries | Naive repair signatures are easy to imitate; the failure map shows which dynamical features an adversary exploits |
| O3. Impostors match living-like systems under every feasible protocol ($V^* \approx ½$) | Recovery dynamics alone are not diagnostic within the tested families; a clear negative result |
| O4. Strong simulation performance, large simulation-to-laboratory gap | Simulated ground truth omits crucial physics; revise testbeds before any mission argument |
| O5. Engineered regenerators are indistinguishable from evolved organisms | Repair behaviour does not encode evolutionary origin; the test detects repair, not Darwinian descent |

**Table 5.** Outcome–interpretation map.

## 6 Discussion

**What the test can and cannot claim.** Repair competence is, we argue, necessary for any system that persists far from equilibrium in a hostile environment, but it is not sufficient evidence of life. Fire, crystals, chemical oscillators and designed regenerators all show recovery of some kind. For this reason the framework treats the abiotic world as an active opponent rather than as a background. Even a strong result would be one rung among several on the Ladder of Life Detection [16], to be combined with independent lines of evidence.

**Definitional dependence.** Class labels rest on a definition of life. We adopt a Darwinian working definition [11] because it is operational, and we place self-replicating abiotic patterns [17] and engineered regenerators [15] in class I to stress-test it. A different definition would move these systems and change what the test measures; we regard making that dependence explicit as a feature.

**False negatives.** Dormant or very slowly metabolizing life (spores, deep-subsurface organisms) may repair on timescales beyond a mission budget. The observation window T and the cost model matter, and the test should be understood as sensitive to *active* repair on accessible timescales.

**Contamination and planetary protection.** A positive interrogation result on a sample taken in situ could reflect terrestrial contamination, so blanks and sterilized controls must be part of any protocol. Destructive perturbation also reduces the material available for confirmatory analysis, which the cost term c(a) must

reflect.

**A speculative connection to ageing theory.** Because repair investment is traded off against other demands [13], the distribution of repair parameters across organisms may itself be informative, and threshold-like transitions between repair-dominated and damage-dominated regimes resemble epidemic thresholds in network models. We note this only as a direction for later work; nothing in the present framework depends on it.

## 7 Limitations

- The admissible family $\Phi$ bounds every claim: a high V* means only that interrogation separates L from impostors *within* $\Phi$. Missing impostor types are missing threats.
- Ground truth in simulation is a modelling decision. Systems labelled living-like are only as life-like as their models.
- The recovery features of Section 3.2 are hypotheses; the learned policy may exploit regularities that do not survive the move to real instruments.
- Perturbation modalities and dose ranges in simulation are analogues; calibrating them against real physical doses is part of Stage A.
- The proposed decision thresholds are provisional and must be fixed before experiments.
- Two cited sources [1, 4] are recent preprints or press-reported work whose details should be verified against the primary documents before submission (marked †).

## 8 Conclusion

We have recast life detection as an adversarial, budgeted interrogation of a sample's response to damage. The value of the associated game is a total-variation measure of how well non-living dynamics can imitate living-like recovery, which turns a vague worry about false positives into a number that can be estimated, compared across impostor families, and reduced by better instruments or better theory. Whether recovery behaviour proves to be a strong, weak or empty signature is an empirical question; the programme above is designed so that any of those answers is a publishable and useful one.

## Declarations

**Use of AI assistance.** Portions of this manuscript were drafted and edited with the assistance of an AI language model (Claude, Anthropic). The author is responsible for all content, claims and citations.

**Funding, competing interests, data and code availability.** To be completed by the author. No data were generated; code for the testbeds is to be released with any empirical follow-up.

## References

[1] Adaptive science operations in deep space missions (2025). arXiv:2510.08812. † [POMDP-based sequencing of the Enceladus Orbilander life-detection suite; confirm authors and version before citing.]

[2] Bartlett, S., Li, J., Gu, L., Sinapayen, L., Fan, S., Natraj, V., Jiang, J. H., Crisp, D., & Yung, Y. L. (2022). Assessing planetary complexity and potential agnostic biosignatures using epsilon machines. *Nature Astronomy*, 6, 387–392.

[3] Benner, S. A. (2017). Detecting Darwinism from molecules in the Enceladus plumes, Jupiter's moons, and other planetary water lagoons. *Astrobiology*, 17.

[4] Can AI detect life? Lessons from artificial life (2026). arXiv:2604.11915. † [Associated with press coverage of adversarial false positives reported at ALIFE 2026; confirm authors and claims before citing.]

[5] Chaloner, K., & Verdinelli, I. (1995). Bayesian experimental design: A review. *Statistical Science*, 10(3), 273–304.

[6] Chan, B. W.-C. (2019). Lenia: Biology of artificial life. *Complex Systems*, 28(3), 251–286.

[7] Dorn, E. D. (2005). *Universal biosignatures for the detection of life*. Ph.D. dissertation, California Institute of Technology.

[8] Dorn, E. D., & Adami, C. (2011). Robust monomer-distribution biosignatures in evolving digital biota. *Astrobiology*, 11.

[9] England, J. L. (2013). Statistical physics of self-replication. *Journal of Chemical Physics*, 139, 121923.

[10] Goodfellow, I. J., Shlens, J., & Szegedy, C. (2015). Explaining and harnessing adversarial examples. *International Conference on Learning Representations*.

[11] Joyce, G. F. (1994). Foreword. In D. W. Deamer & G. R. Fleischaker (Eds.), *Origins of Life: The Central Concepts* (pp. xi–xii). Jones and Bartlett.

[12] Kaelbling, L. P., Littman, M. L., & Cassandra, A. R. (1998). Planning and acting in partially observable stochastic domains. *Artificial Intelligence*, 101, 99–134.

[13] Kirkwood, T. B. L. (1977). Evolution of ageing. *Nature*, 270, 301–304.

[14] Marshall, S. M., et al. (2021). Identifying molecules as biosignatures with assembly theory and mass spectrometry. *Nature Communications*, 12, 3033.

[15] Mordvintsev, A., Randazzo, E., Niklasson, E., & Levin, M. (2020). Growing neural cellular automata. *Distill*.

[16] Neveu, M., Hays, L. E., Voytek, M. A., New, M. H., & Schulte, M. D. (2018). The Ladder of Life Detection. *Astrobiology*, 18(11), 1375–1402.

[17] Pearson, J. E. (1993). Complex patterns in a simple system. *Science*, 261, 189–192.

[18] Salimans, T., Ho, J., Chen, X., Sidor, S., & Sutskever, I. (2017). Evolution strategies as a scalable alternative to reinforcement learning. arXiv:1703.03864.

[19] Scharf, C., et al. (2015). A strategy for origins of life research. *Astrobiology*, 15, 1031–1042.

[20] Schulman, J., Wolski, F., Dhariwal, P., Radford, A., & Klimov, O. (2017). Proximal policy optimization algorithms. arXiv:1707.06347.

[21] Sinapayen, L. (2023). Self-replication, spontaneous mutations, and exponential genetic drift in neural cellular automata. *Proceedings of the Artificial Life Conference (ALIFE 2023)*.

[22] Smith, H. B., & Sinapayen, L. (2024). An agnostic biosignature based on modeling panspermia and terraformation. arXiv:2403.14195; published in *The Astrophysical Journal* (2026), doi:10.3847/1538-4357/ae4ee3.

[23] Sutton, R. S., & Barto, A. G. (2018). *Reinforcement Learning: An Introduction* (2nd ed.). MIT Press.

† Details to be verified against the primary source before submission. References without page numbers were cited at the level of the sources consulted; complete them from the publishers' records.

## Appendix A Adversarial Interrogation Training Loop

**Algorithm 1.** League-style alternating best response

**Input:** living-like sampler *L*; admissible impostor family $\{G_\varphi\}$; budget B; costs $c(\cdot)$; weights $\lambda$, $\beta$; archive *A* initialised with simple impostors

1: Initialise prober parameters $\theta$ (recurrent policy) and generator parameters $\varphi$

2: **for** epoch = 1, …, E **do**

3: **for** k = 1, …, K **do** *// prober update*

4: draw z ~ Bernoulli(½); if z = L sample a system from *L*, otherwise sample an impostor from $G_\varphi$ or from the archive *A*

5: roll out $\pi_\theta$ under budget B, recording $(a_t, y_t, r_t)$ and the terminal decision

6: update $\theta$ by PPO using the shaped reward of Eq. (7) and the terminal reward

7: **end for**

8: sample candidate generators $\varphi_j = \varphi + \sigma\varepsilon_j$ *// evolution strategies*

9: for each j estimate the prober's error on $G_{\varphi_j}$ over M rollouts; subtract an admissibility penalty for violated constraints

10: update $\varphi$ with the ES gradient estimate; every $N_a$ epochs add $G_\varphi$ to *A*

11: **end for**

12: evaluate $\pi_\theta$ against held-out generator families and held-out substrates (Section 4.3); report $A_{rob}$ and accuracy versus budget

**Notes.** (i) Sampling impostors from the archive guards against forgetting earlier attacks. (ii) The generator's admissibility penalty enforces the constraints of Section 3.4, including the absence of heritable variation. (iii) Held-out adversary families are never used for any training or model-selection step.

## Appendix B Reproducibility and Pre-registration Checklist

- Fix decision thresholds for H1–H4 and the budget grid before any run; log all amendments.
- Publish the admissible family $\Phi$, the held-out adversary families and the cost model $c(a)$.
- Report all seeds, including failures, and confidence intervals from at least 20 seeds.
- Include killed-cell and sterilized-sample controls in every wet-laboratory comparison.
- Verify every reference in the bibliography against its primary source, and complete missing page ranges.